Stop Paying Blindly.

Let AI Control the Flow.

CtrlPay — a proactive, defensive personal-finance agent for students

Topic A • Personal Finance Assistant
2026 Shenzhen FinTechathon • International Track

Live: ctrlpay-new-bee4.vercel.app

The problem is not *knowing*. It is *the moment before paying*.

Most budgeting apps are **rear-view mirrors**:

- They show what already left the account.
- They warn after the damage.
- They add another dashboard to ignore.

The student reality

- RM 1,800 monthly income
- Rent + bills + food are fixed
- One RM 180 impulse buy can create a month-end shortfall
- Subscriptions renew silently

> CtrlPay intervenes **before** the money moves.

One loop: receipt → insight → recommendation → action

1. Sense

- OCR a receipt
- Read a payment intent
- Detect a subscription charge
- Advance the sandbox clock

2. Decide

- Deterministic CFaR engine
- Shortfall probability + expected shortfall
- Needs vs wants
- Budget and mandate limits

3. Advise

Groq-hosted GPT-OSS 120B turns **already-computed risk** into plain-

4. Act

Signed T3 contract card:

Approve • Modify • Decline

Live AI, with a hard boundary

Language brain

`app/api/chat` + `lib/groq-client.ts`

- Groq Chat Completions
- `openai/gpt-oss-120b`
- Extracts intent and identifiers
- Writes the explanation

Arithmetic brain

`lib/quant-engine.ts`

- Every RM figure
- Every probability
- Every recommendation threshold
- Pure, deterministic TypeScript

> The model never computes money. The server signs every proposal with HMAC-SHA256.

If the provider is unavailable, the API returns `503` — it never fabricates a "safe" answer.

Real interception, not a mock warning

Payment interceptor

A Shopee **RM 180** want is flagged:

- CFaR and shortfall probability
- Projected month-end balance
- Daily safe allowance
- A separate, live AI advice card

Receipt scanner

The synthetic demo receipt is **LUXE WATCH GALLERY • RM 899**:

- Tesseract extracts the fields
- Groq reviews amount, merchant and budget impact
- A real warning pop-up opens
- The user can edit before logging

Screenshots: `docs/evidence/production-payment-ai-advice.jpeg` • `docs/evidence/production-ocr-ai-warning.jpeg`

Security is the product, not a slide

Authority

- Master Switch **default OFF**
- T0–T3 permission tiers
- T3 requires a per-action click + passkey ref
- Signed, expiring proposals

Evidence

- HMAC-SHA256 proposal signatures
- Hash-chained audit log
- PII redaction before the LLM
- Post-generation RM number verifier
- Strict CSP, no third-party scripts

> The LLM proposes. The engine computes. The human authorizes. The audit chain proves it.

Judge demo in 5 minutes

1. Open the public URL — no login, fresh synthetic student.
2. Trigger **Simulate Shopee RM 180** → deterministic CFaR + live AI advice.
3. Open **Scan receipt** → watch the LUXE WATCH warning pop-up.
4. Approve a T3 contract with a click + passkey → show the signed proposal and audit entry.
5. Turn the Master Switch OFF → repeat T3 → show the **BLOCKED** audit record.

Full script: README.md → “5-minute demo script”

Why this is different

Defensive by design

Not another spend tracker. A **pre-payment interceptor** with deterministic risk and a human decision.

AI where it helps

Language for explanation and extraction. Arithmetic for money. A clean, auditable boundary.

Built for the sandbox

Reproducible clock, seeded simulations, testable engine, labelled ledger buckets.

Built for trust

Every action leaves a reason-coded, hash-chained trail.

What is working today

- Public production deployment: **200 OK**
- Live Groq-backed chat, OCR advisory, payment advice
- Self-hosted Tesseract OCR (no third-party runtime requests)
- 260+ automated tests, strict TypeScript, production build
- Deterministic execution evidence and security self-assessment

> Try it: **ctrlpay-new-bee4.vercel.app**

Source: github.com/zhangze1007/Yi-HUi • Docs: docs/TECHNICAL_DOCUMENTATION.md • SECURITY_REPORT.md

Roadmap after the hackathon

- Real bank/webhook integrations behind explicit consent
- Durable, tenant-scoped proposal store and WORM audit anchoring
- Strong authentication, device binding and step-up biometrics
- On-device OCR and a privacy-preserving model router

Thank you.

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